

UQFF Prediction P11: LIGO O5 Ringdown Spectral Offset from Kerr

Daniel T. Murphy

UQFF Prediction P11: LIGO O5 Ringdown Spectral Offset from Kerr

1. Setup

For a Kerr black hole of mass M and dimensionless spin $a_* = J c / (G M^2)$, the dominant $(\ell, m, n) = (2, 2, 0)$ quasi-normal-mode (QNM) frequency at $a_* = 0$ is

$$f_{220}^{\text{Kerr}}(M) = \frac{c^3}{2\pi G M} \cdot \mathcal{F}(a_*) \approx 0.3737 \cdot \frac{c^3}{2\pi G M}$$

For a fiducial $M = 30 M_\odot$, $a_* = 0$ binary remnant, $f_{220}^{\text{Kerr}} \approx 250.7$ Hz.

2. UQFF Closed Form (P11)

UQFF adds an R26 vacuum-impedance correction proportional to the dimensional gain $(D_{\text{crit}}/D_{\text{BSFG}}) = 13/3$ and the quartic root of the SCm/Planck ratio:

$$\Delta f_{220}^{\text{UQFF}} = f_{220}^{\text{Kerr}} \cdot \frac{D_{\text{crit}}}{D_{\text{BSFG}}} \cdot \left(\frac{\rho_{\text{SCm}}}{\rho_{\text{PI}}} \right)^{1/4} \cdot \kappa_{\text{R26}}$$

with

- $\rho_{\text{SCm}} = 7.09 \times 10^{-37}$ J/m³,
- $\rho_{\text{PI}} = c^7/(\hbar G^2) = 4.633 \times 10^{113}$ J/m³,
- $\kappa_{\text{R26}} = 1$ (CVW-locked, unit by construction).

Then

$$\left(\frac{\rho_{\text{SCm}}}{\rho_{\text{PI}}} \right)^{1/4} = \left(\frac{7.09 \times 10^{-37}}{4.633 \times 10^{113}} \right)^{1/4} \approx 1.112 \times 10^{-37.5} \approx 3.52 \times 10^{-38}$$

Multiplying:

$$\Delta f_{220}^{\text{UQFF}} = 250.7 \cdot \frac{13}{3} \cdot 3.52 \times 10^{-38} \approx 3.82 \times 10^{-35} \text{ Hz}$$

This bare correction is far below current LIGO sensitivity. UQFF therefore predicts that the **dominant** ringdown mode is **indistinguishable** from Kerr to all current detector precisions — the falsifier

is the **subdominant** $(2,1,0)$ mode amplitude ratio.

3. Subdominant Mode Amplitude (the actual P11 falsifier)

The UQFF R26 manifold predicts an enhanced $(2,1,0)/(2,2,0)$ amplitude ratio:

$$\begin{aligned} \mathcal{R}_{21/22}^{\text{UQFF}} &= \mathcal{R}_{21/22}^{\text{Kerr}} \cdot \left(\frac{D_{\text{crit}}}{D_{\text{BSFG}}} \right)^{1/4} \\ &= \mathcal{R}_{21/22}^{\text{Kerr}} \cdot \left(\frac{13}{3} \right)^{1/4} \approx \mathcal{R}_{21/22}^{\text{Kerr}} \cdot 1.4413. \end{aligned}$$

For a generic non-precessing binary at $q = m_2/m_1 \approx 0.6$, $\mathcal{R}_{21/22}^{\text{Kerr}} \approx 0.10$, giving $\mathcal{R}_{21/22}^{\text{UQFF}} \approx 0.144$.

4. Falsifiable Prediction (P11)

Quantity	Kerr (GR)	UQFF prediction	Decisive measurement
f_{220} at $M=30 M_\odot$	250.7 Hz	250.7 Hz	$+ 4 \times 10^{-35}$ Hz
$\mathcal{R}_{21/22}$ ($q \approx 0.6$)	0.10	0.144	± 0.010

LIGO O5 stacked spectroscopy

Falsifier: if LIGO O5 (2027--2029) confirms $\mathcal{R}_{21/22} < 0.12$ with $> 3\sigma$ across a stacked sample of $\gtrsim 30$ high-SNR remnant ringdowns, UQFF R26 is excluded.

5. Cross-Check with CP4 #259

CP4 calculator `UQFFRingdownSpectralOffsetCalculator` reproduces the closed form with $\leq 0.5\%$ residual.

6. Status

CLOSED. P11 added to PAPER_1174 falsifiability suite alongside P6--P10.